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Forecasting Agentic Commerce: How Compute Tokenization Reshapes Transaction Mechanisms in Autonomous Agent-to-Agent Economies

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Abstract

Forecasts of agentic commerce increasingly emphasize the capability trajectory of large AI models, yet realized economic value also depends on whether autonomous agents can discover counterparties, compare offers, authorize transactions, verify performance, and settle payments at machine speed. This conceptual article develops the Compute Tokenization-Transaction Mechanism Transformation (CT-TMT) framework to explain how tokenized compute may become a transaction infrastructure for autonomous agent-to-agent (A2A) economies. Drawing on transaction cost theory, token economics, platform ecosystem theory, and technology-forecasting research, the framework theorizes that compute tokenization operates through three antecedent mechanisms—standardization, financialization, and programmability—that reshape four transaction-mechanism dimensions: price discovery, settlement architecture, trust and verification, and identity and authorization. These mechanisms influence efficiency, market liquidity, resilience, and accountability, while their effects are moderated by technical heterogeneity, regulatory clarity, and market-structure concentration. The article contributes to technology forecasting by treating transaction-mechanism maturity as a mediating variable between AI capability and realized agentic-commerce value. It further extends transaction cost theory to high-frequency algorithmic governance, clarifies compute tokens as capacity-linked productive claims rather than generic crypto-assets, and specifies leading indicators for tracking the 2026-2032 evolution of agentic commerce. A morphological foresight analysis develops three internally consistent scenarios—tokenization-dominant, hybrid equilibrium, and centralized resurgence—and identifies observable signals that can be used in future empirical and Delphi-based forecasting research.

Keywords: agentic commerce; agent-to-agent economy; compute tokenization; transaction mechanisms; technology forecasting; decentralized AI; algorithmic governance

1. Introduction

Autonomous AI agents are moving from passive decision-support tools toward economic actors that can search for resources, evaluate offers, execute payments, and operate within delegated authority envelopes. Industry and academic forecasts describe large agentic-commerce opportunities, but many projections implicitly assume that once AI capability reaches a threshold, economic exchange will follow (McKinsey & Company, 2025; Hadfield & Koh, 2025). That assumption is incomplete. Markets are not created by capability alone. They require transaction mechanisms: price discovery, contracting, settlement, verification, identity, authorization, and accountability.

The emerging agentic economy therefore presents a technology-forecasting problem rather than merely a model-capability problem. A capable agent that cannot authenticate itself, compare the quality of compute

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offers, verify that a remote inference was performed as promised, or settle a machine-readable payment remains economically constrained. Conversely, modest capability gains may translate rapidly into economic value if transaction infrastructure matures. Forecasting agentic commerce therefore requires a framework that connects AI capability with complementary institutional and technical infrastructure.

Recent technical developments make this infrastructural question operationally plausible, although they remain unevenly mature. HTTP-based payment protocols such as x402 define machine-readable payment flows in which a server can return payment instructions and a client can resubmit a request with a signed payment payload (x402 Foundation, 2025). The Linux Foundation announced the x402 Foundation in April 2026 as a neutral home for the protocol after Coinbase contributed the project (Linux Foundation, 2026). Cloudflare's Pay Per Crawl and x402-related integrations illustrate how AI crawlers and agents may encounter paid digital resources through machine-readable interfaces (Cloudflare, 2025a, 2025b). Agent-interoperability protocols such as MCP and A2A, identity proposals such as ERC-8004, and agent-commerce proposals such as ERC-8183 provide additional examples of the emerging transaction stack, but they should be treated as early-stage standards or proposals rather than as evidence of mature market adoption (Anthropic, 2024; Google Cloud, 2025; Ethereum Improvement Proposals, 2025, 2026).

This paper focuses on compute tokenization: the abstraction of computational capacity-such as GPU-hours, inference throughput, proof-generation cycles, or edge-NPU capacity-into standardized, programmable, and potentially tradable units. The paper does not claim that all crypto-assets or all AI payments are compute tokens. Instead, it examines capacity-linked productive claims: digital instruments whose economic value is anchored in access to verifiable computational work and whose transactional usefulness depends on standardization, financialization, and programmability.

The research question is: under what infrastructural and institutional conditions can compute tokenization convert autonomous-agent capability into realized agentic-commerce value? To answer this question, the article develops the Compute Tokenization-Transaction Mechanism Transformation (CT-TMT) framework. The framework integrates transaction cost theory (Coase, 1937; Williamson, 1985), token economics (Catalini & Gans, 2018; Cong, Li, & Wang, 2021; Goldstein, Gupta, & Sverchkov, 2024; Sockin & Xiong, 2023), platform ecosystem theory (Rochet & Tirole, 2003; Parker & Van Alstyne, 2005; Gawer & Cusumano, 2014), and technology-forecasting research (Godet, 2000; Porter et al., 2011; van Notten et al., 2003; Amer, Daim, & Jetter, 2013).

The article makes four contributions. First, it reframes agentic-commerce forecasting by introducing transaction-mechanism maturity as a mediating variable between AI capability and realized economic value. Second, it extends transaction cost theory by specifying algorithmic governance as a high-frequency, low-discretion governance configuration for machine-native exchange, rather than as a simple extension of conventional market contracting. Third, it clarifies compute tokens as capacity-linked productive claims with commodity-like, utility-like, and financialized properties. Fourth, it provides a foresight structure-morphological scenarios and leading indicators-that can guide empirical research, Delphi elicitation, and technology-roadmapping studies of autonomous-agent economies.

The remainder of the paper proceeds as follows. Section 2 reviews the theoretical foundations. Section 3 explains the conceptual and foresight methodology. Section 4 presents the CT-TMT framework. Section 5 develops a stylized analytical representation and operationalization logic. Section 6 derives research propositions. Section 7 develops three scenarios for the 2026-2032 horizon. Sections 8 and 9 discuss implications, limitations, and conclusions.

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2. Theoretical Foundations

The CT-TMT framework draws on four theoretical traditions. Transaction cost theory explains why exchange governance changes when search, negotiation, monitoring, enforcement, and settlement costs change. Token economics explains how cryptographic tokens can coordinate network participation and economic incentives. Platform ecosystem theory explains how multi-sided infrastructures govern interactions among heterogeneous participants. Technology-forecasting and foresight research explains how emerging technical artifacts can be organized into future-oriented scenarios under uncertainty.

2.1 Transaction cost theory and algorithmic governance

Transaction cost theory (TCT) begins from the premise that firms and markets are alternative governance responses to the costs of transacting (Coase, 1937; Williamson, 1985). Classic TCT emphasizes bounded rationality, opportunism, asset specificity, uncertainty, and transaction frequency. Digital markets have historically reduced search and negotiation costs, while platforms and smart contracts have reduced some contracting and enforcement frictions (Bakos, 1997; Bajari & Tadelis, 2001; Werbach & Cornell, 2017; Savelyev, 2017).

Autonomous A2A exchange requires a careful extension of TCT. AI agents are not boundedly rational in Simon's human cognitive sense; their behavior is bounded by training data, architecture, objective functions, tool permissions, and authority envelopes (Simon, 1957). Opportunism is not necessarily moral opportunism; it may manifest as objective misalignment, adversarial optimization, unverifiable output quality, or delegated action beyond an authorized boundary. In this context, algorithmic governance can be understood as a governance configuration in which predefined protocols, cryptographic proofs, escrow, slashing, and machine-readable authorization replace repeated human negotiation and ex-post dispute resolution.

The claim is not that algorithmic governance eliminates transaction costs. Rather, it shifts the cost structure. It raises fixed protocol-design and compliance costs but lowers marginal negotiation, settlement, and enforcement costs when transaction frequency is high. This makes algorithmic governance especially relevant for per-call inference, pay-per-crawl access, real-time data retrieval, and other high-frequency microtransactions that are poorly served by conventional procurement or subscription billing.

2.2 Token economics and capacity-linked productive claims

Token economics examines how tokens function as media of exchange, incentive devices, governance instruments, and claims on network-specific utility (Catalini & Gans, 2018; Cong, Li, & Wang, 2021; Sockin & Xiong, 2023). Existing token categories include utility tokens, security tokens, stablecoins, governance tokens, and commodity-linked tokens. Compute tokenization overlaps with these categories but should not be collapsed into any one of them.

A more precise typology is required. Payment tokens are used to settle compute transactions but do not themselves represent computational capacity. Utility compute tokens grant access to a platform-specific compute network. Capacity-backed compute claims represent rights to measurable computational work, such as GPU-hours, inference throughput, or proof-generation cycles. Staking and governance tokens secure or govern compute networks but may only indirectly represent productive capacity. This article

uses the term compute tokenization primarily to refer to capacity-linked productive claims and the transaction mechanisms that make such claims comparable, verifiable, tradable, and programmable.

This clarification matters because the economic value of capacity-linked compute claims is partially anchored in physical productive capacity. Unlike pure utility tokens, their valuation may be bounded by the marginal cost and availability of equivalent centralized compute. Unlike stablecoins, their price need not be pegged. Unlike security tokens, they need not confer ownership or cash-flow rights. Their distinctive property is that they can serve as both a transactional instrument and an input into further autonomous production.

2.3 Platform ecosystem theory and autonomous multi-sided markets

Platform ecosystem theory examines how platforms enable interactions among distinct user groups, generate network effects, and govern complementor participation (Rochet & Tirole, 2003; Parker & Van Alstyne, 2005; Eisenmann, Parker, & Van Alstyne, 2006; Boudreau, 2010; Gawer & Cusumano, 2014). Conventional platforms are governed by human organizations that design pricing rules, access controls, complementor policies, and dispute procedures. Blockchain-mediated platforms and decentralized autonomous organizations have already complicated this view by shifting some governance functions into protocols (Lumineau, Wang, & Schilke, 2021).

Autonomous compute markets extend the platform problem further. Demand agents, supply agents, verification agents, governance agents, and payment facilitators may all be software actors operating at machine-native frequency. Platform governance therefore becomes less about managing human complementors and more about maintaining protocol neutrality, preventing cryptographic attacks, managing token-economic instability, and ensuring auditable delegation. This configuration is an autonomous multi-sided market: a market infrastructure in which interaction sides are partly or wholly non-human and governance is executed through a mixture of code, standards, registries, and institutional rules.

2.4 Technology forecasting and socio-technical infrastructure

Technology-forecasting research emphasizes that emerging technologies do not diffuse automatically from technical capability to economic value. Forecasts should consider complementary assets, institutional readiness, regulatory conditions, infrastructure, market structure, and social acceptance (Porter et al., 2011; Geels, 2002; Saritas & Aylen, 2010; Amer, Daim, & Jetter, 2013). Scenario methods are especially appropriate when a domain has high uncertainty, path dependency, and interacting technical and institutional drivers (van Notten et al., 2003; Godet, 2000).

Agentic commerce is such a domain. The relevant uncertainty is not merely whether AI models become more capable. It is whether transaction infrastructure can mature in parallel with capability. Compute tokenization may accelerate machine-native markets, but its effects depend on standards, payment interoperability, verification mechanisms, regulatory treatment, and compute-market concentration. The CT-TMT framework therefore treats transaction mechanisms as socio-technical mediators rather than neutral technical plumbing.

3. Methodological Approach: Conceptual Theory Building and Artifact-Based Foresight

This paper develops a conceptual framework using a theory-building logic appropriate for domains in which empirical regularities are emerging but not yet mature enough for large-sample estimation (MacInnis, 2011; Jaakkola, 2020). The approach combines artifact-based mapping with foresight-oriented abstraction. The aim is not to catalogue all AI-agent or blockchain projects, but to identify the mechanisms through which observable technical artifacts may transform transaction governance.

The procedure involved four steps. First, relevant artifacts were identified when they met at least one of four criteria: a public protocol or standard specification; a peer-reviewed article or technical preprint with a stable identifier; production documentation from a major infrastructure provider; or a widely referenced implementation directly relevant to payment, identity, verification, benchmarking, or compute coordination. Illustrative artifacts included MCP for model-tool interoperability, A2A for agent interoperability, x402 for HTTP-native payment, ERC-8004 for agent identity/reputation/validation, ERC-8183 for proposed escrow-based agentic commerce, MLPerf Endpoints for API-centric benchmarking, and OpenID Foundation work on agentic identity management (Anthropic, 2024; Google Cloud, 2025; x402 Foundation, 2025; Ethereum Improvement Proposals, 2025, 2026; MLCommons, 2026; OpenID Foundation, 2025). Second, artifacts were grouped by functional role and abstracted into higher-order constructs. Third, the constructs were interpreted through TCT, token economics, platform ecosystem theory, and technology-forecasting literature. Fourth, a morphological foresight analysis translated the framework into scenario configurations for the 2026-2032 horizon.

Artifact class	Illustrative examples	Constructs informed
Agent interoperability and identity protocols	A2A communication protocols, Model Context Protocol, ERC-8004-style identity/reputation/validation registries, decentralized identifiers	Identity and authorization; accountability; platform governance
Machine-readable payment protocols	x402, pay-per-crawl integrations, micropayment channels, escrow-based job protocols	Settlement architecture; programmability; efficiency
Compute benchmarking and abstraction	MLPerf Endpoints, workload-specific benchmarks, vNPU virtualization, FLOPs/token or latency-throughput metrics	Standardization; price discovery; technical heterogeneity
Verification and attestation mechanisms	zkML proofs, trusted execution environments, optimistic fraud proofs, evaluator attestations	Trust and verification; resilience; accountability
Decentralized compute and staking systems	Compute marketplaces, staking/slashing models, proof-of-useful-work proposals, dual-token architectures	Financialization; market liquidity; market-structure concentration

Table 1. Artifact-to-construct mapping used in the CT-TMT framework.

The analysis deliberately distinguishes artifact evidence from causal evidence. Emerging protocols and industry implementations show that a transaction-mechanism problem is becoming observable; they do not by themselves prove that tokenized compute markets will dominate. The propositions and scenarios developed below are therefore framed as conditional claims and as a research agenda for future empirical validation.

4. The CT-TMT Framework

The Compute Tokenization-Transaction Mechanism Transformation framework has four layers: antecedent mechanisms, transaction-mechanism dimensions, outcome categories, and moderators. The

framework's core claim is that compute tokenization affects agentic-commerce outcomes through transaction mechanisms rather than through compute supply alone.

Layer	Construct	Definition and operational meaning
Antecedent	Standardization	Conversion of heterogeneous compute into comparable units of account through benchmarks, workload metrics, and hardware abstraction. Operational indicators include benchmark adoption, comparable latency-throughput reporting, and cross-provider performance disclosure.
Antecedent	Financialization	Transformation of compute capacity into a tradable, stakeable, collateralizable, or yield-bearing asset. Operational indicators include staking depth, token-market liquidity, collateral use, and slashing events.
Antecedent	Programmability	Embedding compute transactions in machine-readable protocols, smart contracts, escrows, and conditional settlement logic. Operational indicators include programmable payment transactions, escrowed jobs, and automated settlement volume.
Transaction mechanism	Price discovery	Process through which agents arrive at comparable clearing prices for compute. Indicators include bid-ask spread, price dispersion for similar workloads, auction participation, and price-response speed.
Transaction mechanism	Settlement architecture	Infrastructure for finalizing compute transactions and payments. Indicators include settlement latency, finality certainty, failed-payment rate, and cross-platform settlement interoperability.
Transaction mechanism	Trust and verification	Mechanisms that verify output quality, execution integrity, or provider claims. Indicators include zk/TEE attestations, evaluator validation, proof-verification cost, and dispute rate.
Transaction mechanism	Identity and authorization	Mechanisms that establish agent identity, reputation, authority boundaries, and permissioned delegation. Indicators include registered agent identities, delegated authority scopes, compliance attestations, and identity portability.
Outcome	Efficiency	Reduction in search, negotiation, verification, settlement, and enforcement frictions.
Outcome	Market liquidity	Depth, continuity, and geographic breadth of compute supply and demand.
Outcome	Resilience	Fault tolerance, censorship resistance, and robustness to provider failure or platform lock-in.
Outcome	Accountability	Traceability, auditability, and liability assignment for autonomous transactions.
Moderator	Technical heterogeneity	Degree to which hardware, workloads, chains, benchmarks, and APIs remain fragmented.
Moderator	Regulatory clarity	Degree to which jurisdictions define agent authority, liability, audit requirements, and payment compliance.
Moderator	Market structure concentration	Degree to which compute supply is controlled by hyperscalers or a distributed provider base.

Table 2. CT-TMT constructs and operational meaning.

4.1 Antecedent mechanisms

4.1.1 Standardization

Standardization converts heterogeneous computational resources into comparable units. Compute hardware differs across GPUs, NPUs, ASICs, memory bandwidth, interconnects, geographic location, energy cost, and workload-specific performance. Without standardization, autonomous agents cannot reliably compare offers or use compute as a meaningful unit of account (Jevons, 1875). Benchmarking efforts such as API-centric inference evaluation, latency-throughput reporting, and verified operating points can reduce information asymmetry and enable price comparison, a mechanism consistent with information-economics arguments about quality uncertainty (Akerlof, 1970). MLPerf Endpoints is one

recent example of API-centric benchmarking for generative AI services (MLCommons, 2026). In CT-TMT, standardization primarily affects price discovery and trust because it makes claims about compute quality comparable and auditable.

4.1.2 Financialization

Financialization turns compute capacity from an operational resource into an asset that can be traded, staked, collateralized, or used to secure network participation. This does not imply that every compute token is a financial security. Rather, financialization refers to the economic mechanisms through which compute claims gain liquidity, collateral value, and incentive-alignment properties. Staking and slashing may discipline provider behavior, while liquid markets may broaden access to geographically distributed supply. The relationship is conditional: volatile tokens, thin liquidity, or speculative yield can destabilize rather than deepen compute markets.

4.1.3 Programmability

Programmability embeds compute transactions in executable rules. Machine-readable payment protocols, escrowed job markets, automated evaluator attestations, and smart-contract settlement allow agents to transact without repeated human intervention. x402 illustrates a machine-readable payment flow for API access and digital services, while ERC-8183 proposes an escrow-and-evaluator-attestation primitive for job-based agent commerce; Vaziry, Garzon, and Küpper (2025) provide an early technical architecture combining A2A, ledger-anchored identities, and x402 micropayments. Programmability primarily affects settlement architecture, trust verification, and authorization because it makes conditional exchange executable: pay if and when a provider delivers a validated output under an authorized scope.

4.2 Transaction-mechanism dimensions

The four transaction-mechanism dimensions are the mediating core of the framework. Price discovery affects whether agents can identify efficient providers and avoid quality-adjusted price dispersion. Settlement architecture affects whether high-frequency and granular transactions can be finalized at acceptable cost and risk. Trust and verification affect whether agents can rely on remote computation without direct observation. Identity and authorization affect whether agents can act within traceable permission boundaries and whether responsibility can be assigned after autonomous action; this problem is also central to emerging agentic-identity work by the OpenID Foundation and to ERC-8004-style identity, reputation, and validation registries (OpenID Foundation, 2025; Ethereum Improvement Proposals, 2025).

4.3 Outcomes and moderators

The outcome categories are intentionally broader than short-term transaction volume. Efficiency captures reductions in search, negotiation, settlement, and enforcement friction. Market liquidity captures depth and continuity of compute supply and demand. Resilience captures the ability of the system to withstand outages, censorship, attacks, or provider concentration. Accountability captures auditable records, compliance verification, and liability traceability. Three moderators define boundary conditions. Technical heterogeneity limits comparability. Regulatory clarity shapes adoption risk and compliance cost. Market-structure concentration determines whether decentralized supply can discipline incumbents or remains residual.

5. Analytical Representation and Operationalization

This section provides a stylized analytical representation of the CT-TMT framework. The purpose is not to estimate a structural model in this article, but to clarify causal logic and identify measurable variables for future empirical work.

5.1 Transaction cost decomposition

For an A2A compute transaction T , total transaction cost can be represented as:

$$TC(T) = c_{search} + c_{negotiate} + c_{verify} + c_{settle} + c_{enforce}$$

Tokenized A2A governance changes the relative magnitude of each component. Search costs may fall through registries and standardized capability discovery. Negotiation costs may fall through auctions or algorithmic bargaining. Verification costs may shift from ex-post audit to proof generation and verification. Settlement costs may fall through programmable payment channels but may rise when gas fees or compliance requirements increase. Enforcement costs may fall through escrow and slashing but may rise when smart-contract exploits or oracle dependence require remediation.

5.2 Compute-token valuation as a capacity-linked claim

A capacity-linked compute claim can be represented as a function of three value components and one net risk adjustment:

$$p_{compute}(t) = \alpha(t)MC(t) + \beta(t)U(t) + \gamma(t)Y(t) - \rho(t)D(t)$$

where $MC(t)$ is the marginal cost of equivalent compute provisioning, $U(t)$ is platform-specific utility, $Y(t)$ is staking or yield-related value, and $D(t)$ is a net discount for volatility, regulatory uncertainty, settlement failure, fraud risk, gas-fee exposure, and other transaction risks. The component weights satisfy $\alpha(t) + \beta(t) + \gamma(t) = 1$, with $\alpha(t)$, $\beta(t)$, and $\gamma(t)$ non-negative; $\rho(t)$ is a non-negative sensitivity to the risk-discount term. The weights are not assumed to be stable. In immature markets, utility expectations and speculative yield may dominate. In mature markets, prices should increasingly anchor to quality-adjusted compute cost. The subtraction of $\rho(t)D(t)$ prevents the model from treating tokenized compute as frictionless and highlights why speculative, operational, or regulatory risk can offset transaction-mechanism gains.

5.3 Agentic-commerce value

The central forecasting relation can be summarized as:

$$ACV(t) = K(t) \times \tau(t) \times \delta(t)$$

where $ACV(t)$ is realized agentic-commerce value, $K(t)$ is the capability-linked opportunity ceiling, $\tau(t)$ is transaction-mechanism efficiency, and $\delta(t)$ is adoption diffusion. Capability-only forecasts implicitly assume that $\tau(t)$ and $\delta(t)$ are close to one. CT-TMT instead treats them as endogenous variables shaped by standards, settlement protocols, verification infrastructure, identity systems, regulation, and market structure.

5.4 Algorithmic governance crossover

Algorithmic governance becomes attractive when the fixed cost of protocol deployment is outweighed by reductions in marginal transaction cost. A simplified crossover expression is:

$$N^* = (K_{\text{protocol}} - F_{\text{market}}) / [(c_{\text{neg}} + c_{\text{set}} + c_{\text{enf}}) - \varepsilon_{\text{protocol}}]$$

where N^* is the transaction-frequency threshold, K_{protocol} is the fixed cost of protocol design and integration, F_{market} is the fixed cost of conventional market governance, c_{neg} , c_{set} , and c_{enf} are marginal negotiation, settlement, and enforcement costs under conventional governance, and $\varepsilon_{\text{protocol}}$ is the marginal protocol cost. The expression is meaningful only when $K_{\text{protocol}} > F_{\text{market}}$ and $(c_{\text{neg}} + c_{\text{set}} + c_{\text{enf}}) > \varepsilon_{\text{protocol}}$; otherwise there is no positive crossover threshold under the assumed cost ordering. It should be read as a heuristic rather than a universal law. It clarifies why tokenized A2A exchange is most plausible in high-frequency, low-discretion, machine-executable transactions.

Proposition	Model component	Possible empirical indicators
P1a-P1b	Standardization -> price discovery -> liquidity	Benchmark adoption; price dispersion; bid-ask spread; number of comparable offers per workload
P2a-P2b	Financialization -> settlement/liquidity, moderated by volatility	Staking depth; slashing events; liquidity depth; token volatility; collateral usage
P3a-P3b	Programmability -> transaction-cost reduction, frequency dependent	Programmable payment volume; escrowed jobs; settlement latency; cost per transaction; failure rate
P4	Verification/identity mediate accountability	Attestation count; validation-registry usage; dispute rate; audit-trail completeness
P5-P7	Moderation by heterogeneity, regulation, and market structure	Hardware fragmentation index; regulatory clarity index; hyperscaler share; decentralized supply share
P8	$ACV = K \times \tau \times \delta$	Capability benchmarks; transaction-mechanism maturity index; adoption diffusion; realized agentic transaction volume

Table 3. Proposition-model-measurement mapping.

6. Research Propositions

6.1 Standardization, price discovery, and liquidity

P1a. Higher adoption of compute-performance standards is negatively associated with quality-adjusted price dispersion across comparable workloads.

P1b. The relationship between compute-performance standardization and market liquidity is mediated by price-discovery efficiency.

The logic follows information economics. A market cannot clear efficiently when agents cannot compare the quality of offers. Standardized benchmarking reduces asymmetric information and allows agents to express bids over comparable units. In compute markets, this is likely to reduce bid-ask spreads and increase provider participation, but only where benchmarks cover the relevant hardware and workload classes.

6.2 Financialization, settlement, and liquidity stability

P2a. Staking, slashing, and collateralization mechanisms are positively associated with settlement reliability and provider commitment in tokenized compute markets.

P2b. The liquidity benefits of compute-token financialization are weaker when token volatility and speculative yield dominate capacity-linked valuation.

Financialization can deepen markets by transforming compute supply into tradable or collateralizable claims. However, it can also introduce instability if token prices become detached from compute capacity. This proposition therefore treats financialization as ambivalent: it can improve liquidity, but only when incentive mechanisms remain anchored to productive capacity.

6.3 Programmability and high-frequency transaction efficiency

P3a. Programmable payment and escrow protocols reduce marginal negotiation, settlement, and enforcement costs in A2A compute transactions.

P3b. The efficiency gains from programmability increase with transaction frequency and decrease with protocol-integration cost, gas-fee volatility, and compliance overhead.

Programmability is most valuable when transactions are granular and frequent. Per-call inference, crawler payments, and micro-procurement of digital services are more likely to cross the algorithmic-governance threshold than infrequent enterprise contracts.

6.4 Trust, verification, identity, and accountability

P4. The effect of compute tokenization on accountability is mediated by cryptographic verification mechanisms and portable agent identity infrastructure, with stronger mediation in regulated or high-value contexts.

Without verification and identity, tokenized compute markets face a lemons problem: low-quality or unverifiable providers can undercut high-quality providers. zkML proofs, TEEs, evaluator attestations, reputation registries, and delegated authority envelopes are therefore not peripheral features. They are transaction-mechanism infrastructure.

6.5 Boundary conditions

P5. Technical heterogeneity weakens the relationship between standardization and price-discovery efficiency when benchmarks do not cover relevant hardware, workloads, or settlement networks.

P6. Regulatory clarity strengthens the relationship between programmability and realized transaction outcomes by reducing legal uncertainty around agent authority, liability, auditability, and payment compliance.

P7. Compute-market concentration weakens the relationship between financialization and market liquidity when decentralized supply remains residual relative to hyperscaler-controlled capacity.

6.6 Integrative forecasting proposition

P8a. Capability-only forecasts overestimate realized agentic-commerce value when transaction-mechanism efficiency or adoption diffusion remains low.

P8b. Capability-only forecasts underestimate realized agentic-commerce value when transaction-mechanism efficiency and adoption diffusion improve nonlinearly through open standards, regulatory clarity, and interoperable settlement infrastructure.

7. Foresight Analysis: Scenarios for 2026-2032

The 2026-2032 horizon is appropriate because the relevant technologies are no longer purely speculative, but their institutional and market trajectories remain unsettled. This section uses morphological scenario construction rather than point prediction. The scenarios are internally consistent configurations of key uncertainties, not claims of calibrated probability.

7.1 Key uncertainties and morphological field

Key uncertainty	Low-friction state	Intermediate state	High-friction state
Benchmark standardization	Broad adoption of open, workload-sensitive benchmarks	Sector-specific or platform-specific benchmarks	Proprietary metrics and weak comparability
Payment interoperability	Open machine-readable protocols widely integrated	Partial adoption across selected ecosystems	Closed billing APIs and platform-specific payment rails
Verification maturity	Routine zk/TEE/evaluator attestations for high-value workloads	Selective verification in regulated or premium segments	Provider self-reporting and weak auditability
Agent identity and authorization	Portable identities, reputation, and delegated authority envelopes	Limited ecosystem-level identity portability	Platform-specific identities and ambiguous delegation
Regulatory clarity	Clear authority, liability, audit, and payment rules across major jurisdictions	Fragmented regional clarity	Restrictive backlash or persistent ambiguity
Compute-market structure	Distributed supply meaningfully constrains incumbents	Coexistence of decentralized and hyperscaler supply	Hyperscaler lock-in and residual decentralized supply

Table 4. Morphological field for agentic-commerce infrastructure scenarios.

7.2 Scenario 1: Tokenization-dominant machine-native markets

In the tokenization-dominant scenario, open benchmarking becomes the common language of compute negotiation, machine-readable payment protocols are widely integrated, and verification becomes routine for high-value computation. Portable agent identity and authorization registries allow cross-platform delegation, while major jurisdictions converge on workable authority-envelope frameworks. Decentralized compute supply does not eliminate hyperscalers, but it becomes large enough to discipline pricing and provide geographic redundancy.

Under this scenario, $\tau(t)$ approaches a high level because price discovery, settlement, verification, and identity all mature together. Adoption diffusion accelerates as agents can transact across platforms without bespoke integrations. The primary risks are security incidents, speculative token instability, and regulatory fragmentation, but the system's open standards reduce dependence on any single provider.

7.3 Scenario 2: Hybrid equilibrium

The hybrid equilibrium scenario is the central path of incremental institutional change. Some sectors adopt tokenized compute for high-frequency, low-value, or cross-border transactions, while enterprise workloads with high compliance requirements remain on centralized cloud contracts. Benchmarks advance unevenly across workload classes. Payment protocols are used in selected ecosystems, but large platforms maintain proprietary billing APIs. Verification is used for regulated or high-value workloads, while lower-risk transactions rely on reputation and provider self-reporting.

In this scenario, $\tau(t)$ reaches an intermediate level and $\delta(t)$ diffuses unevenly across sectors and jurisdictions. Agentic commerce grows, but it does not become a single open machine-native market. Instead, it becomes a layered ecosystem in which centralized and tokenized transaction mechanisms coexist.

7.4 Scenario 3: Centralized resurgence

In the centralized-resurgence scenario, one or more boundary conditions fails. A major exploit, fraud event, regulatory backlash, or sustained token-market instability discourages open tokenized infrastructure. Hyperscalers retain control over the most reliable compute capacity, and autonomous agents transact primarily through closed APIs, enterprise procurement systems, or platform-specific wallets. Benchmarks remain proprietary, identity is not portable, and verification is used mainly inside centralized trust domains.

Here, AI capability $K(t)$ may continue to improve, but $\tau(t)$ and $\delta(t)$ remain low for open agentic commerce. This scenario illustrates the core CT-TMT forecasting claim: capability advancement alone is insufficient when transaction infrastructure fails to mature.

7.5 Leading indicators for scenario tracking

Indicator	Tokenization-dominant signal	Hybrid signal	Centralized-resurgence signal
Benchmark adoption	Open benchmarks used in procurement and protocol pricing	Benchmarks used in selected sectors	Proprietary provider metrics dominate
Payment interoperability	x402-like protocols integrated across agents, APIs, and providers	Limited use in specific ecosystems	Closed billing APIs dominate
Verification infrastructure	zk/TEE/evaluator attestations become routine for high-value jobs	Verification limited to regulated or premium workloads	Provider self-reporting dominates
Agent identity	Portable agent identity and reputation registries gain cross-platform usage	Identity portability within selected networks	Platform-specific identity silos
Regulatory environment	Authority envelopes and liability rules clarified in major jurisdictions	Regional fragmentation	Restrictive backlash or persistent ambiguity
Market structure	Distributed supply constrains incumbent pricing	Centralized and decentralized supply coexist	Hyperscaler concentration remains dominant
Token-market behavior	Capacity-linked valuation and deep liquidity	Liquidity in selected tokens and use cases	Speculative volatility overwhelms capacity linkage

Table 5. Leading indicators for tracking CT-TMT scenarios.

8. Discussion

8.1 Theoretical implications

The first implication concerns TCT. CT-TMT does not argue that autonomous agents make governance irrelevant. It argues that machine-native exchange changes the cost structure of governance. Algorithmic governance is most defensible as a high-frequency, low-discretion governance configuration in which protocols substitute for repeated negotiation and some forms of ex-post enforcement. This narrows and strengthens the theoretical claim relative to a broad assertion that smart contracts constitute an entirely new institutional order.

The second implication concerns token economics. Compute tokenization should be analyzed through a typology of claims rather than through a generic crypto-token category. Payment tokens, utility tokens, capacity-backed claims, and governance/staking tokens have different valuation logics and regulatory implications. The productive character of capacity-linked compute claims makes them distinct because their value is partially anchored in physical computational work.

The third implication concerns platform ecosystem theory. Autonomous multi-sided markets extend platform research by making non-human participation and cryptographic governance central rather than peripheral. The relevant governance problems shift from complementor pricing and platform access alone to verification, delegation, protocol neutrality, and auditable accountability.

8.2 Forecasting implications

For technology forecasting, the article's central implication is that AI capability should not be forecast in isolation from transaction infrastructure. The relation $ACV(t) = K(t) \times \tau(t) \times \delta(t)$ identifies capability, transaction-mechanism maturity, and adoption diffusion as jointly necessary. Forecasting models that condition only on model benchmarks risk overstating realized value when settlement, verification, identity, or regulation lag. They also risk understating value when open standards and transaction infrastructure create nonlinear diffusion.

The framework also encourages jurisdiction-sensitive forecasting. Regulatory clarity is not a background variable; it can determine whether autonomous agents are permitted to bind principals, settle payments, and produce auditable records. Geographic variation in authority envelopes, payments regulation, and AI liability can therefore produce uneven diffusion even when technical protocols are globally available.

8.3 Practical implications

For platform designers, the framework suggests an architectural sequence. Standardization should precede deep financialization because liquid markets require comparable units. Verification and identity should develop alongside programmability because autonomous settlement without accountability can generate adverse selection. For policymakers, the framework suggests that workable authority envelopes and audit requirements may enable safer autonomous commerce than blanket restrictions or ambiguous liability. For investors and corporate strategists, the framework identifies leading indicators-benchmark adoption, protocol integrations, verification use, identity portability, regulatory clarity, and market concentration-that should be monitored alongside AI capability metrics.

8.4 Social-change implications

Agentic commerce may reshape more than payment flows. It can alter procurement labor, platform power, access to compute, compliance practices, and the geography of digital infrastructure. If tokenized compute broadens access to distributed capacity, it may reduce dependence on a small number of cloud providers. If it remains speculative or captured by incumbents, it may reinforce concentration. The environmental implications are also ambiguous: finer-grained markets may improve utilization of idle compute, but financialized demand can also increase energy-intensive computation. These social-change effects should be incorporated into future empirical research rather than treated as secondary consequences.

9. Limitations and Future Research

This article has limitations. First, it is a conceptual and foresight paper rather than an empirical test. The propositions require validation using on-chain data, protocol adoption records, benchmark diffusion, provider pricing, regulatory variation, and transaction-level settlement data. Second, the analytical model abstracts from real-world frictions such as gas-fee volatility, maximal extractable value, oracle failure, smart-contract exploits, and strategic behavior by heterogeneous agents. Third, the focus on compute tokenization leaves data tokenization and model tokenization for future work. Fourth, the scenarios are morphological configurations rather than calibrated probability forecasts. Delphi studies, prediction markets, expert elicitation, and cross-impact analysis could refine their relative plausibility.

Future research can proceed in four directions. Empirical studies can test whether benchmark adoption reduces price dispersion in compute markets. Event studies can examine whether staking or slashing mechanisms affect liquidity and provider reliability. Comparative institutional research can analyze how regulatory clarity affects adoption across jurisdictions. Agent-based simulations can model how autonomous buyers and sellers interact under different verification, identity, and settlement regimes.

10. Conclusion

The agentic economy is not only a story about more capable AI models. It is also a story about whether autonomous agents can transact through reliable, auditable, and programmable infrastructure. This article developed the CT-TMT framework to explain how compute tokenization may reshape the transaction mechanisms required for A2A economies. Standardization, financialization, and programmability can transform price discovery, settlement architecture, trust and verification, and identity and authorization. These mechanisms affect efficiency, liquidity, resilience, and accountability, but only under favorable boundary conditions of technical interoperability, regulatory clarity, and market-structure openness.

The central forecasting claim is conditional: realized agentic-commerce value depends on capability, transaction-mechanism maturity, and adoption diffusion. Compute tokenization may become a machine-native substrate for autonomous exchange, but it may also remain a niche infrastructure or be absorbed into centralized platform systems. By specifying mechanisms, propositions, operational indicators, and scenarios, the CT-TMT framework provides a foundation for future empirical and foresight research on the economic infrastructure of autonomous AI agents.

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